

1 · What we did with the microphones: substitution calibration

SoundVisualizer · chamber report, 2026-10-05 · calibrator session 2026-09-16

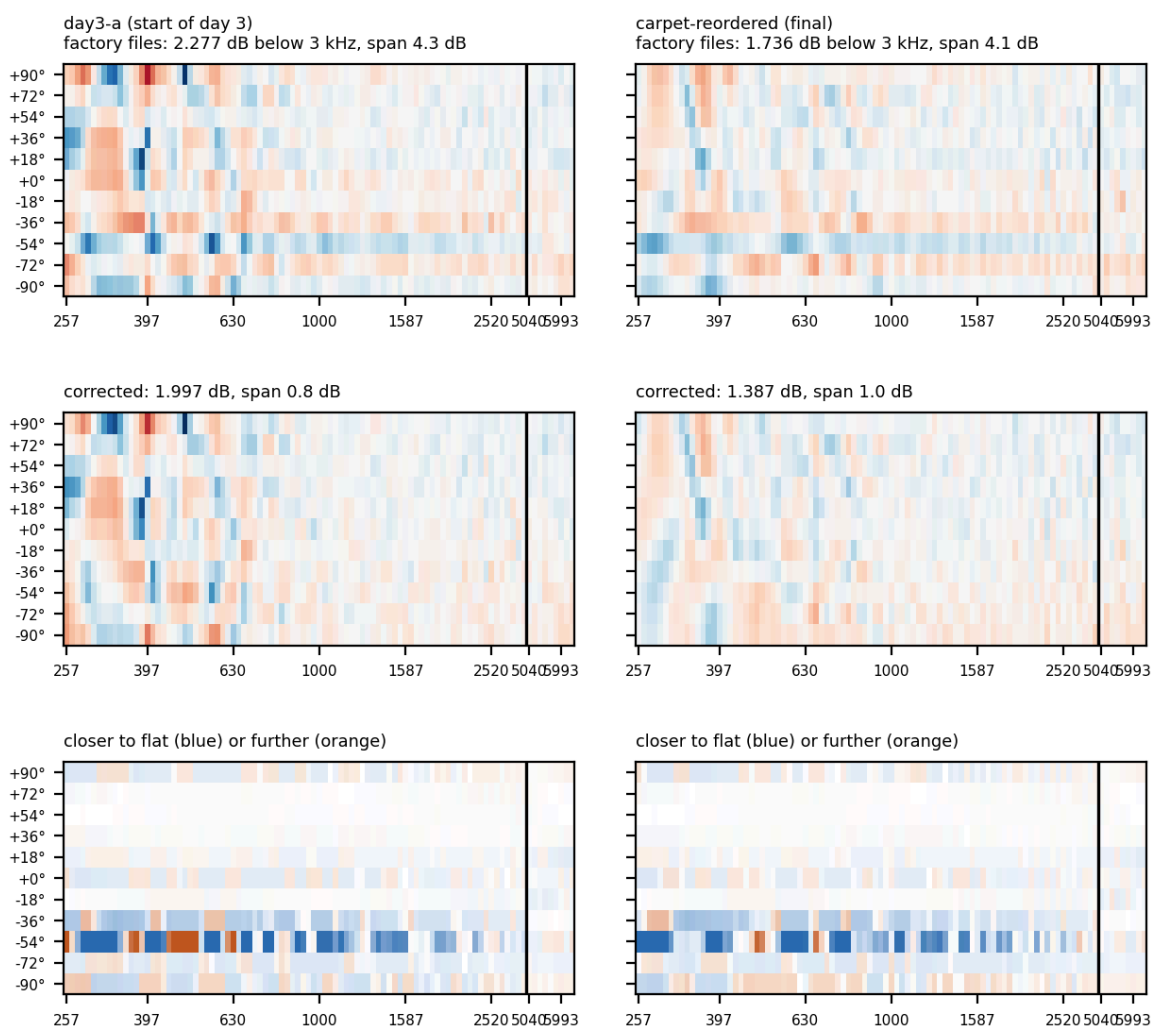


The source: a printed 1 l sphere with an 8 cm driver, at the hub.

Eleven UMIK-2 capsules sit on the 1.68 m ring. Their factory calibration files were never checked against each other, so on 16 Sep every capsule was measured against every other at one seat (55 pairs, 12 tones per octave). At one seat the source and the room are the same for both capsules, so their difference is the capsule.

- **Seven of the eleven factory files were wrong by 0.6–3.6 dB**, clustered by serial prefix.
- The corrections went into each capsule's calibration curve. The spread across the eleven, for the same sound, fell from **4.01 dB to 0.02 dB**. Repeatability of the method: 0.08 dB sd.
- The datum is the four capsules whose files agree with measurement. Still open: the absolute level, which needs a 94 dB calibrator.
- On the 30 Sep prop-plane polar the corrections cut the roughness from 3.3 to 1.0 dB.

Before and after, on two captures from this report



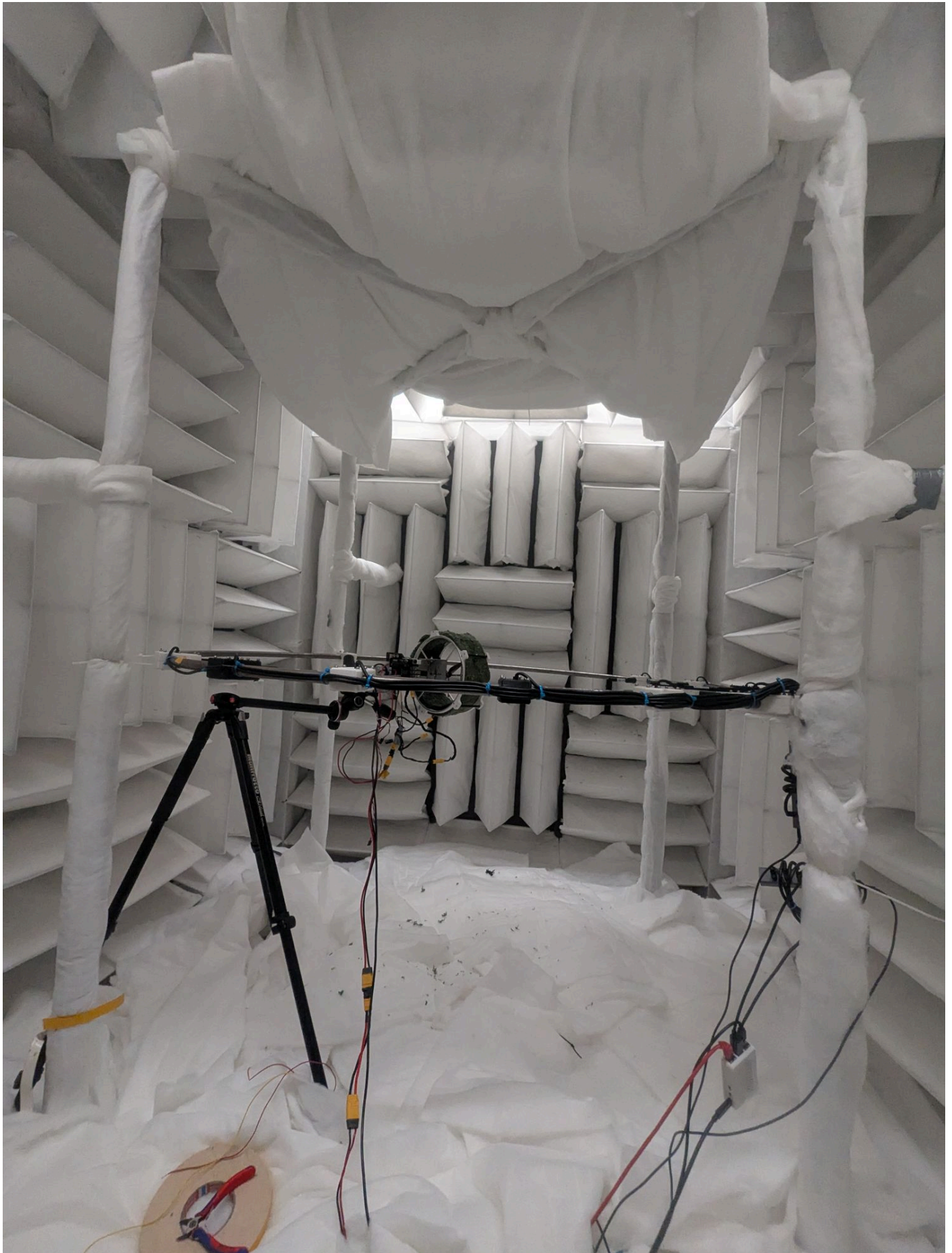
Hz (3–5 kHz omitted). Rows: level of each capsule relative to the arc mean (± 12 dB); bottom row ± 3 dB. Top of each map = $+90^\circ$.

Same two captures, calibrated two ways: the factory files (top) and the measured corrections (middle); “span” is the range of the capsules' mean levels. The factory files add a different offset to each capsule, which shows up as horizontal stripes across every frequency; the span of the capsules' mean levels is **4.3 $\rightarrow$ 0.8 dB** at the start of day 3 and **4.1 $\rightarrow$ 1.0 dB** in the final configuration. Room error below 3 kHz changes **2.277 $\rightarrow$ 1.997 dB** and **1.736 $\rightarrow$ 1.387 dB**. The change is a per-capsule offset, so it moves rows, not the frequency structure of the room; every room-error number elsewhere in this report uses the corrected files.

Numbers from the project notes (CLAUDE.md, calibrator session 2026-09-16), which govern where this page and they differ.

2 · Main waterfalls and the day 3 evaluation

2.1 the chamber



The chamber in the final configuration (carpet - reordered, photographed 5 Oct): the arc ring laid flat around the propeller rig at the hub, the speaker tripod on the left, batting on the floor and ceiling, wedges and lined pillars on the walls. The waterfall on the next page compares the start of day 3 with this state.

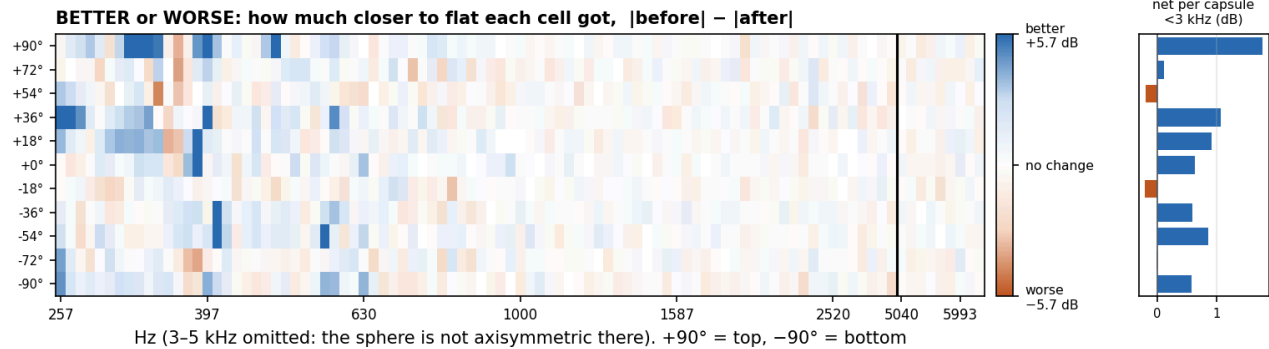
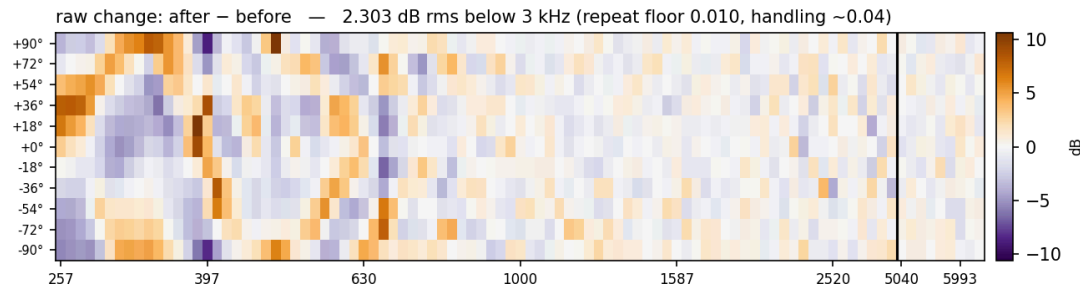
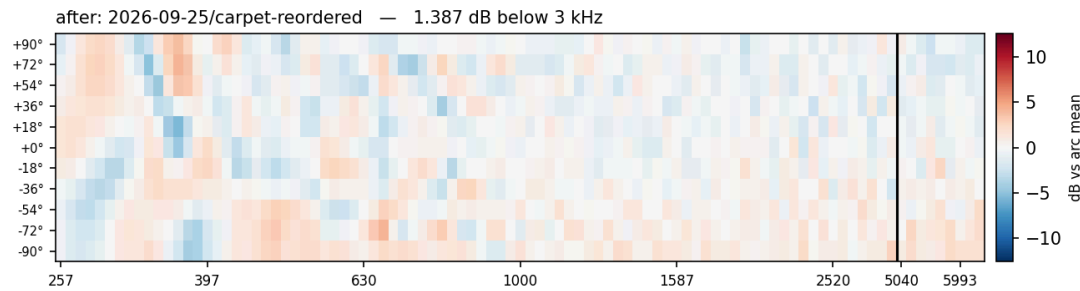
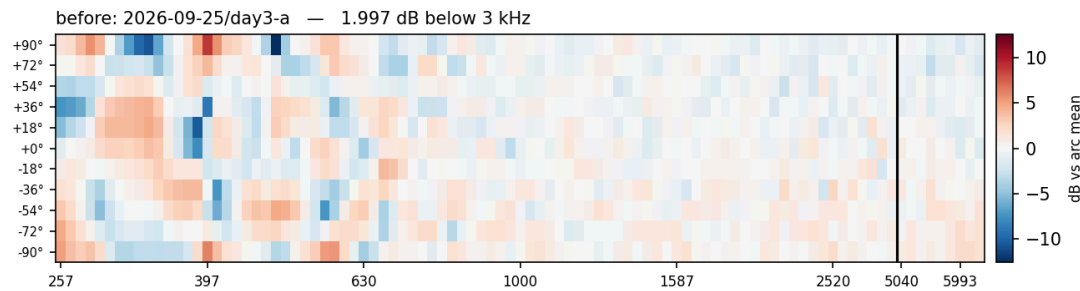
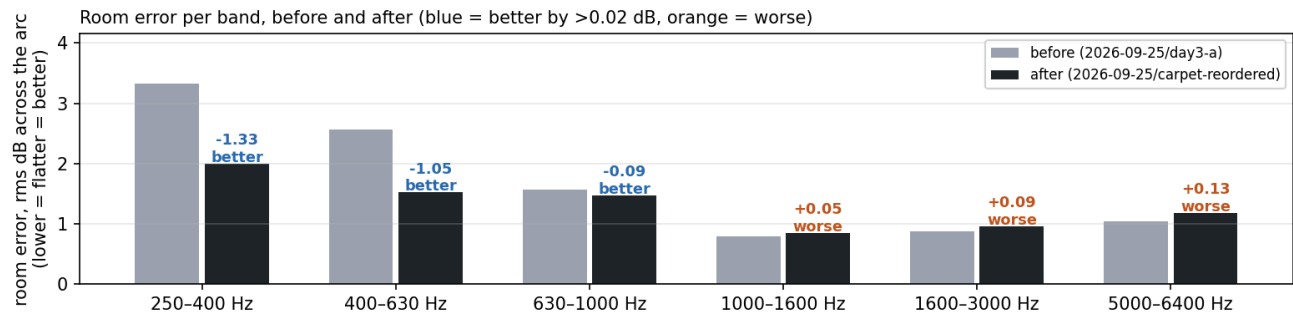
2 · Main waterfalls and the day 3 evaluation

2.2 main waterfall: day 3 start against the final state

Both runs are from 25 Sep with the corrected microphones. The source was moved between them (page 4).

Day 3 start (worst) vs final · source moved in between: room error below 3 kHz 1.997 → 1.387 dB (-0.610, better)

2026-09-25/day3-a → 2026-09-25/carpet-reordered · below 3 kHz, 492 of 946 cells moved >0.05 dB toward flat and 415 away from it · 95 tones · 11 calibrated capsules



2 · Main waterfalls and the day 3 evaluation

2.3 evaluation against the ISO anechoic tolerance values (an analogue, not a qualification)
11 calibrated capsules on a 0.84 m arc · 95 tones, 257 Hz–6.35 kHz · runs: 25 Sep, day3-a (13:17) → carpet-reordered (19:15)

The method

Where the method comes from. Taken from the standards: each one-third-octave band judged on its own against the free-field ideal, tones and noise separately, the worst position against the limit, and the anechoic tolerances ± 1.5 dB for 125–630 Hz and ± 1.0 dB for 800–5000 Hz (ISO 5305:2024 Table 1, p. 8; the same values in Cunefare et al. 2003, J. Acoust. Soc. Am. 113(2), p. 882, quoted on page 5; ISO 3745:2012 and ISO 26101 are the two procedures, compared by Russo et al. 2018). Ours, not the standards: they move one microphone along a traverse and test the decay of level with distance, and only that test earns the words "qualified" or "in conformity" (ISO 26101-1 p. 3; ISO 3745 §5.1). We have eleven fixed capsules at one radius and test uniformity around an axisymmetric source, so the reference is the arc mean and the same values are applied to a different quantity. **Band level** = each capsule's mean over the 8–9 tones in the band, worst capsule against the value; **pure tones** = share of tone × capsule cells inside; within ± 0.03 dB is marginal (largest difference between two same-day runs of one unchanged state, 5 pairs). The room is treated and not qualified, so this is a figure of merit that ranks bands; it claims neither qualification nor conformity. Held: previews of ISO 3745:2012 (clauses 1–6.1.2, no annexes), ISO 26101-1 and ISO 5305.

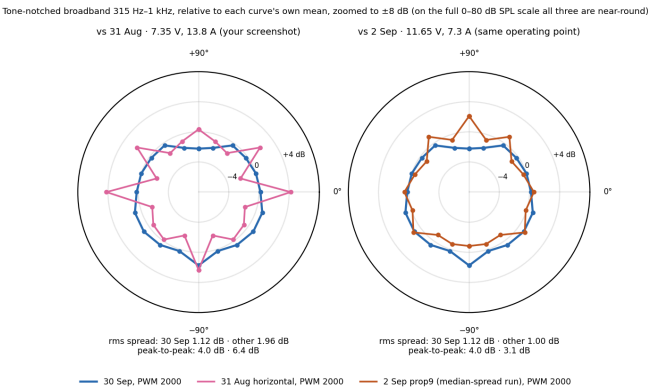
Cell = one tone × one microphone. "Cells in limit" counts these, e.g. 250 Hz: 3 tones × 11 microphones = 33 cells, 12 inside ± 1.5 dB = 36 %. The deviation columns show the worst microphone's band mean, as a share of the limit.

BAND	LIMIT	TONES (× 11 CAPSULES)	DAY 3 START	CELLS IN LIMIT	FINAL	CELLS IN LIMIT
250 Hz	± 1.5	3	6.81	36 %	1.88	52 %
315 Hz	± 1.5	8	3.46	28 %	0.89	56 %
400 Hz	± 1.5	8	3.46	41 %	1.41	59 %
500 Hz	± 1.5	8	1.93	41 %	1.99	59 %
630 Hz	± 1.5	8	0.93	43 %	1.49	61 %
800 Hz	± 1	8	0.87	60 %	1.12	55 %
1000 Hz	± 1	9	1.08	71 %	0.57	73 %
1250 Hz	± 1	8	0.54	83 %	1.02	73 %
1600 Hz	± 1	8	0.69	75 %	1.15	76 %
2000 Hz	± 1	8	0.66	84 %	0.77	82 %
2500 Hz	± 1	8	0.82	64 %	0.75	65 %

MEASURE	DAY 3 START	FINAL
Room error below 3 kHz (dB)	1.997	1.387
Within the tolerance values from (band)	1250 Hz	2000 Hz
Bands clearly over the limit (of 11)	5: 250, 315, 400, 500, 1000 Hz	4: 250, 500, 800, 1600 Hz
Bands marginal	0: – Hz	2: 630, 1250 Hz
Tone × capsule cells inside the limit	58 %	65 %

Colour = worst-capsule deviation as a share of the band's limit: green 0, yellow at the limit, red at twice the limit or more (cells in limit: green 100 %, yellow 50 %, red 0 %). The cut-off is set by the highest failing band, so read it with the lists. **Caveat:** the source was moved between these runs (20 cm closer at 13:44, into the ring plane at 14:03): the map changed by 2.30 dB rms, against 0.14 for a re-arranged carpet and 0.80 for a 5 cm move, so the gain is treatments and source positions together. The one pair with an unmoved source is 24 Sep, empty floor 2.054 → carpet 1.317 (three arrangements 1.317 / 1.318 / 1.324).

3 · Polars: is the prop-plane polar rounder than before?



TONE-NOTCHED BROADBAND, 315 HZ–8 KHZ, CELLS WITHIN ± 1.3 DB OF THE POLAR MEAN	RUNS	PWM 2000: MEDIAN (RANGE)	WORST CELL	PWM 1900: MEDIAN (RANGE)
30 Sep, today	1	66 %	3.1 dB	72 %
2 Sep, same operating point (11.65 V, 7.3 A, –4.0 N)	10	71 % (65–79)	3.4 dB	76 % (65–84)
31 Aug (7.35 V, 13.8 A, +6 N: other operating point)	5	65 % (38–68)	7.5 dB	62 % (38–66)

- Rounder than 31 Aug where it counts:** worst capsule 3.1 dB off the mean instead of 7.5 dB; the 31 Aug runs go down to 38 % of cells in tolerance. That run is the one in the screenshot, whose total level (tones included) is equally spread, 1.39 vs 1.41 dB; the gain is in the broadband.
- Not rounder than 2 Sep** (same voltage, current, thrust): 66 % against a median of 71 % (65–79), inside the scatter of the 2 Sep runs.
- Prop-plane measurement:** ends of the arc read louder than the centre at 2 and 4 kHz by +2.3 and +2.2 dB today, and by +1.5 and +4.3 dB (median) in the flat-arc runs where the true difference is zero: position-fixed error, not directivity.

4 · Semi-anechoic room solutions that apply to this chamber

What fails (accepted configuration, evaluated on its own): clearly over the ISO limit at 250, 500, 800, 1600 Hz, marginal at 630, 1250 Hz; the blade tone sits at 215–258 Hz and its third harmonic at 713 Hz. Nearest reflectors measured: extra path 0.68 m at -72° and 1.49 m at $+36^\circ/+54^\circ$, i.e. 1.98 and 4.34 ms. Source: chamber-fighting-guide.pdf §03, 29 facilities.

SOLUTION	WHAT THE PAPERS GIVE	APPLIES TO US?	WHAT WE ALREADY KNOW
1. Move the geometry (source, arc, reflector)	Sets the ceiling for everything else. Reaching 500 Hz needs every reflector ≥ 0.69 m of extra path, 250 Hz needs 1.37 m (gate ≥ 5 ms, Matelján, ARTA note 4).	Yes, first, free. The 0.68 m surface is just short for 500 Hz.	Source position was the largest lever in our trials: moving the speaker about 20 cm closer took the room error from 1.96 to 1.31 dB within one afternoon (25 Sep).
2. Measured room correction per position and tone	Source measured once in a real chamber and again in the room, band by band: "the uncertainty in measurements is minimal (below 0.55 dB for frequencies above 200 Hz)" (du Plessis et al. 2022, p. 29).	Yes, the strongest. Valid for a fixed source and fixed microphones — our arc — not for a flying drone.	Needs a reference run in a chamber (the university's 300 m ³). Void if a capsule is re-seated: 1 cm moves the response ± 6 dB at high frequency (Bellmann & Klippel, p. 7). Must be checked against the prop at 200–250 Hz (Mehrgou 2012, p. 34).
3. Average over source speed	Tones are where a modal room hurts most; a pattern that changes over a 19 % frequency step averages out over a speed ladder. No published residual.	Yes, cheap. Hardware exists; 12–15 PWM steps.	Tone-pattern holes of 8 dB moved with a 19 % frequency step in the flat-arc runs; the polar shape of 30 Sep depends on speed outside 1.3–2 kHz.
4. Time gating	Lowest usable frequency ≈ 1 / gate length: "the time-bandwidth requirement is satisfied on frequencies above 177.9 Hz" for a 5.6 ms gate (Matelján, p. 8).	Only above 500 Hz: our gate is 1.98 ms.	Good for naming the 0.68 m and 1.49 m reflectors; cannot reach the blade tone.
5. Absorb: foam, wedges	34 cm of depth per 250 Hz; cost about $\times 4$ per octave down (Orrego et al. 2018, p. 477). Removing the floor treatment costs ~ 3 dB broadband and up to 10 dB at the blade tone (Ma et al. 2022, p. 7).	Not at 250 Hz. Yes at 630–1000 Hz: about 10 cm on the named surface.	The floor under the arc and source mattered most (wedges off: 2.05 $\rightarrow$ carpet 1.32, same afternoon); the same material on the walls did nothing.
6. Subwavelength absorber	"99.2% absorbance at 239 Hz in experiment" in 100 mm (Long et al. 2020, Sci. Rep. 10:13823).	The one exception for the blade tone, but only if the 238 Hz error is the room, not the stand.	Decide with the reverse-rotation capture first.

Order for this chamber

- Record the room (hub height, dimensions, what stands within 1.5 m of the arc ends) and move the 0.68 m reflector or the arc away from it.
- One capture with the propeller turning the other way, everything else identical: separates stand-induced from room-induced asymmetry.
- Loudspeaker correction at the blade harmonics (215–258 Hz, 713 Hz) for the eleven positions, then compare with the propeller at the same positions.
- Speed ladder to average the tones, then treat only the surfaces the sweep names (about 10 cm of foam at 630–1000 Hz).

Microphone placement against the standards

- Meets:** far field ($(kr)^2 \approx 15$ at 250 Hz, above 3); the ring radius 0.84 m is more than $\lambda/4$ from 250 Hz up; capsule axis normal to the measurement surface (ISO 3745 §6.1.1); ISO 5305 R $\geq 5 \cdot D_A$ holds for the 137 mm sphere and one 6-inch propeller, not for larger rotors.
- Does not meet:** UMIK-2 is not a WS2F/WS3F class 1 free-field microphone (ISO 5305 §5.1, ISO 3745 §6.1.1); the ring, clamps and cables are untreated supports; one radius only, so decay with distance is not tested.
- Not recorded:** distance of the ring ends to the walls and floor, hub height, ISO 5305 H_m .

"TABLE I. Maximum allowable difference in anechoic rooms between measured and theoretical free-field levels per ISO 3745 and ANSI S12.35."

Cunefare et al. 2003, J. Acoust. Soc. Am. 113(2), p. 882

Where the accepted configuration stands, and what the method cannot say

- Evaluated on its own (25 Sep): room error 1.387 dB, within the strict anechoic tolerance values from the 2000 Hz band up, 65 % of tone cells inside. **Comparable rooms in the held papers** (not the 63–275 Hz rotor chambers, which are better rooms): a 6.8 m³ low-cost box conforming to the ISO 3745 anechoic row from 500 Hz with one point 1.0 dB over (Orrego et al. 2018, p. 6); a 51 m³ room that passes with noise and fails with pure tones (Nash 2019); Amazon Prime Air's rotor room, ± 1 dB only above $2.2 \times \text{BPF}$ and ± 4 dB below (Nardari 2019, p. 2). Rooms tested against the standard keep the strict row and publish the range that passes (Russo 2018: "in conformity" for a reduced range); rooms that cannot meet it use their own tolerance. We keep the strict row, as Orrego, Nash and Kayhan did, and report the failures. No held paper judges a room by the angular spread across a fixed ring.
- Deviation from the arc mean is blind to an error shared by all eleven positions; the ISO traverse has not been run, so this ranks bands and does not certify. The anechoic table is used because the floor under the arc is absorbing.
- Open, unexplained: a two-state switch in the source chain.** The arc-wide balance of 5–6.4 kHz against 257–400 Hz sits on two levels and flips by the same amount between neighbouring runs: foam-2 $\rightarrow$ foam-out -4.3, ceiling-carpet $\rightarrow$ ceiling-carpet-day2 (next morning) +4.4, ceiling1-floor2 $\rightarrow$ felt-floor-only -4.1, chaotic-carpet-2 $\rightarrow$ -3 +4.3, in-plane-b $\rightarrow$ curtain +4.6 dB. Speaker device and amplitude were unchanged at the same-day flips. Not a source move (a 5 cm move changes the map by 0.80 dB; the 24 Sep flip by 0.12), not redistributed absorber (room-only changes that afternoon moved the mean level by at most 0.13 dB), not a gain or supply-voltage change (that would shift every frequency alike; 257–400 Hz does not move). It cancels in the arc-relative score. To check: cable near the sphere, connectors, amplifier and supply (300 Hz and 5 kHz on live_tone while handling each).
- The grid starts at 257 Hz; 3–5 kHz is excluded (sphere), 5–6.4 kHz indicative.